

CSE260 Assignment 01

This assignment must be hand-written. Show ALL steps in ALL questions.

1.[5 marks] Convert the following binary number to equivalent decimal number

a) $(11011.101)_2$

b) $(0.0001)_2$

2.[5 marks] Perform the following base conversions

a) $(29)_{12} = (?)_7$

b) $(10110111)_5 = (?)_4$

3. [5 marks] Perform addition, subtraction and multiplication for the pair of following base-9 numbers. Verify your results by converting the problem into decimal.

412

134

4. [5 marks] Perform the following arithmetic operations using 15-bit two's complement and one's complement systems . State if there is an overflow in each case.

a) $379 + 98$

5. [5 marks] You want to buy two 8 GB DDR4 RAMs. Each RAM costs $(1C2)_{16}$ dollars. You also want to buy a graphics card RTX4070Ti which costs $(10010110000)_2$ dollars. However, you don't have that much money with you. Suddenly, one of your generous friends agreed to give you the money you need. He decided to give you $(4064)_8$ dollars. How much will you have left after buying those components? (Show the answer in decimal)